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```

Experiment 9 : Global Thresholding

```
clc; close all; clear;
```

Global Thresholding

```
I = imread('Images10/septagon_orig.tif');
Inoisy = imread('Images10/septagon_noisy.tif');
Ishade = imread('Images10/septagon_shade.tif');
Ithumb = imread('Images10/thumb_noisy.tif');

HistOrig = imhist(I);
HistNoisy = imhist(Inoisy);
HistShade = imhist(Ishade);
HistThumb = imhist(Ithumb);

% Thresholds
T = graythresh(I);
Tnoisy = graythresh(Inoisy);
Tshade = graythresh(Ishade);
Tthumb = graythresh(Ithumb);

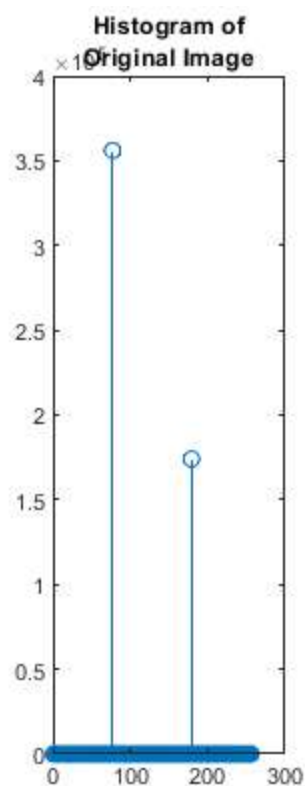
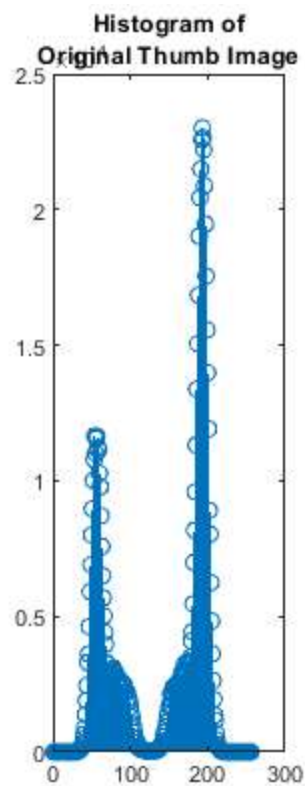
G = imbinarize(I, T);
Gnoisy = imbinarize(Inoisy, Tnoisy);
Gshade = imbinarize(Ishade, Tshade);
Gthumb = imbinarize(Ithumb, Tthumb);

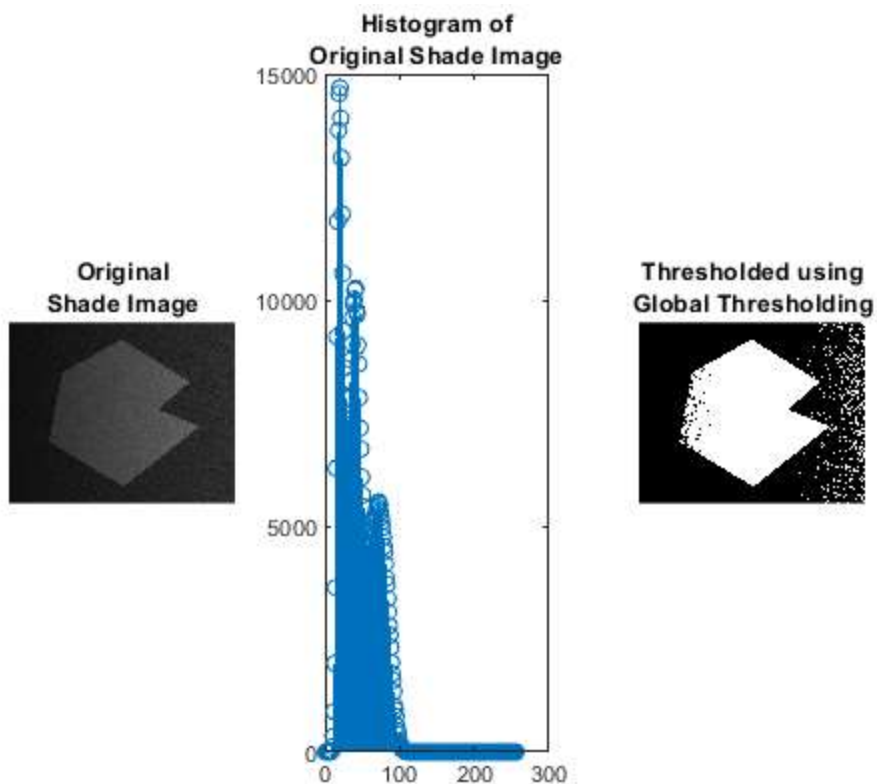
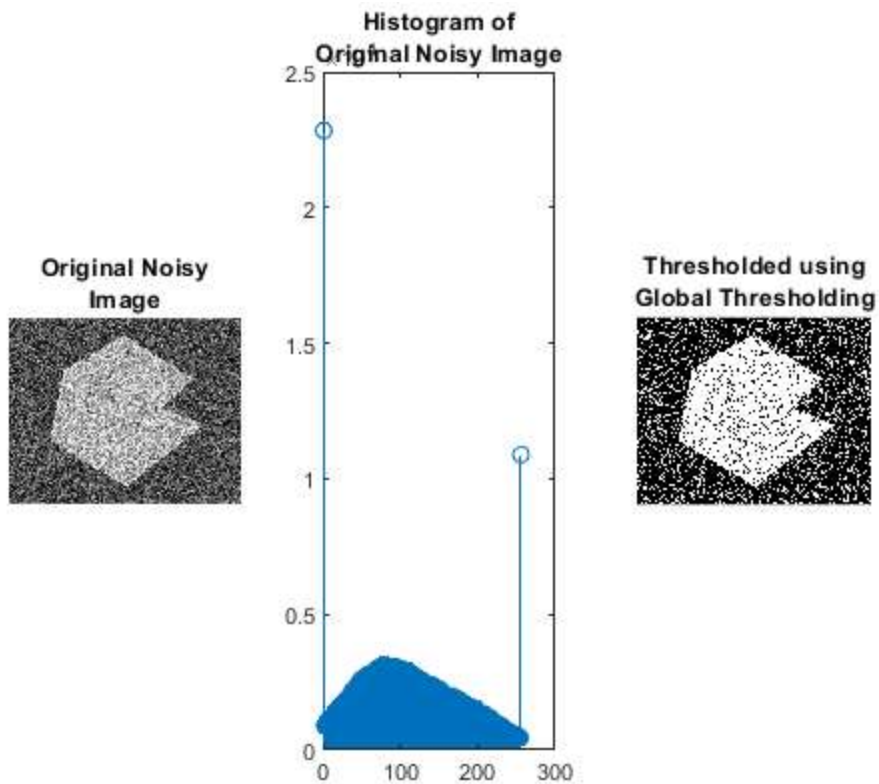
figure,
subplot(1, 3, 1), imshow(Ithumb), title({'Original',"Thumb Image"});
subplot(1, 3, 2), stem(HistThumb), title({'Histogram of', "Original Thumb Image"});
subplot(1, 3, 3), imshow(Gthumb), title({'Thresholded using', "Global Thresholding"});

figure,
subplot(1, 3, 1), imshow(I), title("Original Image");
subplot(1, 3, 2), stem(HistOrig), title({'Histogram of', "Original Image"});
subplot(1, 3, 3), imshow(G), title({'Thresholded using', "Global Thresholding"});

figure,
subplot(1, 3, 1), imshow(Inoisy), title({'Original Noisy', "Image"});
```

```
subplot(1, 3, 2), stem(HistNoisy), title({"Histogram of", "Original Noisy Image"});  
subplot(1, 3, 3), imshow(Gnoisy), title({"Thresholded using", "Global Thresholding"});  
  
figure,  
subplot(1, 3, 1), imshow(Ishade), title({"Original", "Shade Image"});  
subplot(1, 3, 2), stem(HistShade), title({"Histogram of", "Original Shade Image"});  
subplot(1, 3, 3), imshow(Gshade), title({"Thresholded using", "Global Thresholding"});
```





Conclusion

Noise and noiseless images were thresholded using Global Thresholding successfully. The images having Bi-Modal histograms have resulted in well thresholded images. On the other hand, images having Uni-Modal and Multi-Modal histograms have noise and have lost some object parts (information) in the output images.

